
title: Project Milestone 2
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High level description
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Add a high level description of your project here. Make sure to summarize all the features that you have designed.

The project includes the beginning stages of implementing a multi-threading library for xv6. We incorporated two sys calls, `create_thread` and `join_thread`. So far, we have implemented `create_thread` to where it can take multiple arguments. It can maintain multiple threads executing at once.

Thread creation
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Describe the decisions you have made for thread create, including answers to the questions above.

1. We decided to let the user allocate the memory for the stack of a thread, that way a user has more control over the memory space of the thread(not static basically).
2. We allow our user to pass the `thread_id` (really the pid of the process) to `join_thread` as well as the address of a status variable to get the status returned by copyout in `join_thread`.

Stack separation
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Describe how your threads gets assigned different stacks (even though they are still in separate address spaces for now).

The user allocates memory and passes the pointer through the `create_thread` function. The function then takes the pointer and maps

it to the trapframe of the respective thread.

Test cases

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- Describe the test case you used for the creation of the threads.

We developed a test code called `thread_test`, in which creates six threads, half of which perform a `doanything1` function and the other does a `doanything2` function. `Doanything1` contains a random sleep time period to prove the threads run concurrently. Both functions also use the input argument from creating the thread to identify what thread is working, which shows that our arguments work. The test case then joins the threads and prints out the pid of the thread as well as the returned status

Addition comments

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Add any addition comments, questions, or design decisions that you are considering for the final submission.